ReThread — Product Requirements Document

Product	ReThread — Clothing Exchange & Swap Marketplace
Programme	Unified Mentor · Web Development Internship
Author	Lakshay Pandit
Version / date	1.0 · October 2026
Status	Phase 1 built, tested and live
Live site	https://rethread-ten.vercel.app

1. Summary

ReThread is a web marketplace where people **swap clothes instead of buying new ones**. Members list garments they no longer wear, discover items from people nearby, send swap requests (one-to-three items per side), negotiate through counter-offers and chat, and complete the exchange through a local meetup or a tracked courier. A transparent **Swap Value Calculator** keeps trades fair, and **location-based matching** surfaces good swaps within a few kilometres. An **admin panel** gives moderators analytics, user and listing management, swap monitoring and dispute resolution.

2. Background & problem

Fast fashion has sharply increased clothing consumption. India alone accumulates up to **7,800 kilotonnes of textile waste a year — about 8.5% of the global total — and roughly half of it is post-consumer** (Fashion for Good, *Wealth in Waste*, 2022). Many wearable garments sit unused because selling or donating them is inconvenient.

Problem (from the brief)	How ReThread addresses it
People have unused clothes but no easy way to exchange them	Two-minute listing with photo compression and auto-valuation
E-commerce focuses only on buying and selling	Item-for-item barter; no payments, prices or wallets anywhere
Resale pricing is complicated	Rule-based value estimate with a visible breakdown and a ±25% fair band
Few platforms are dedicated to swapping	Swap-first flows: request, counter-offer, agreement, fulfilment, rating
Sustainable options are limited	Nearby matching, meetups, and impact tracking (kg kept in use, CO_2e, water)

2.1 Competitive research

Platform	Model	What we borrow	Gap ReThread fills
Depop	Social resale app	Profile-as-closet, likes, direct messages	Money-based; no barter
Poshmark	Resale + offers	Offer/counter-offer negotiation	Pricing friction; seller fees
Swap.com	Consignment resale	Condition grading, catalogue filters	Not peer-to-peer swapping
This For That (Gurugram, 2019)	Credit-based swapping, curation team	Indian context, quality checks, courier slabs	Credits act like money; no local matching
Vinted	Peer-to-peer resale	Size/brand filters, buyer protection	Swap feature limited; no fairness engine

Insight: no mainstream product combines (a) direct item-for-item swaps, (b) an objective fairness signal, and (c) neighbourhood matching — the three things that make swapping feel safe and worthwhile.

3. Goals & objectives

Primary (from the brief) 1. A marketplace dedicated to clothing exchange. 2. Direct swaps without monetary transactions. 3. Encourage sustainable fashion practices. 4. Location-based swap matching.

Secondary 5. Negotiation before finalising swaps (counter-offers + chat). 6. Estimated swap value calculation. 7. Filters by category and location. 8. Courier support for remote swaps (partner + tracking + cost estimate). 9. Community around sustainable fashion (profiles, reviews, impact).

Non-goals (Phase 1, as per brief): online payments, AI recommendations, AR try-on, native mobile apps.

4. Users & personas

Persona	Needs	Key features
Aanya, 20, DU student (Delhi) — budget-conscious, loves thrifting	Refresh wardrobe for free; meet near metro	Browse + distance filter, fair matches, meetup planner
Kabir, 29, professional — wedding-season outfits worn once	Rehome expensive ethnic wear fairly	Value calculator, counter-offers, ratings
Priya, 34, parent (Dwarka) — kids outgrow clothes fast	Swap kidswear bundles locally	Kids category, nearby suggestions
Meera, 31, frequent traveller	Remote swaps	Courier option, tracking numbers

Persona	Needs	Key features
Platform admin	Keep the community safe and measure growth	Analytics, moderation, disputes, exports

5. Scope (Phase 1)

In scope and implemented: registration/login · profiles · clothing listings · browsing & filters · swap requests · negotiation (counter-offers + chat) · swap value calculator · location-based suggestions & map · fulfilment (meetup / courier) · ratings · reports · disputes · notifications · admin panel with analytics and CSV reports.

6. Functional requirements

6.1 User module

ID	Requirement	Status
U1	Register with name, email, password (≥ 8 chars, letter + number), neighbourhood (list or GPS)	✅
U2	Log in / log out; session in httpOnly cookie (7 days)	✅
U3	Profile: name, bio, private phone, sizes (top/waist/footwear), wishlist categories, location	✅
U4	Public profile: closet, swaps completed, kg kept in use, reviews	✅
U5	Wishlist (save items)	✅
U6	Swap history with statuses	✅

6.2 Clothing listing module

ID	Requirement	Status
L1	Upload 1–5 photos (JPG/PNG/WebP, compressed in browser to ≤ 1200 px, server checks magic bytes, 2 MB cap)	✅
L2	Details: title, category (23), gender, size (context-aware), brand, condition (4 levels), colour, material, age, MRP, description	✅
L3	Estimated swap value + owner adjustment within ±25%	✅
L4	Availability status: available → reserved → swapped (or removed)	✅
L5	Edit / remove own listings (blocked while in an agreed swap)	✅
L6	“Looking for” categories to improve matching	✅

6.3 Swap request module

ID	Requirement	Status
S1	Request 1–3 items from one member, offering 1–3 of your own	✅

ID	Requirement	Status
S2	Incoming / sent / history views with “your move” indicators	✅
S3	Accept, decline (with reason), or counter-offer (change items on either side); up to 12 rounds	✅
S4	Accepting = swap agreement; all items reserved atomically; other open requests for those items auto-closed	✅
S5	Fulfilment: meetup (place + future date/time) or courier (partner + AWB, tracking link, cost estimate)	✅
S6	Both members confirm receipt → completed; items marked swapped; stats & impact updated	✅
S7	Cancel before items move; dispute after agreement	✅
S8	Ratings (1–5 stars + comment), one per member per swap	✅
S9	Timeline of every status change	✅

6.4 Negotiation chat module

ID	Requirement	Status
C1	One thread per pair of members, started from a listing, profile or swap	✅
C2	Offer cards and system messages for every swap event inside the chat	✅
C3	Real-time delivery & typing indicator (Socket.io); automatic polling fallback on serverless hosts	✅
C4	Unread counts (navbar badge + per thread), read receipts on open	✅
C5	Swap context bar with status and “View swap” link	✅
C6	Safety nudge if someone mentions UPI/bank details	✅

6.5 Swap value calculator

ID	Requirement	Status
V1	Estimate from brand, condition, category (+ optional MRP and age)	✅
V2	Show value breakdown	✅
V3	Suggest fair matches (single items or pairs from your closet)	✅
V4	Two-sided value comparison with fairness meter	✅
V5	Public calculator page and `POST /api/valuation/estimate`	✅

6.6 Location-based matching

ID	Requirement	Status
G1	Show nearby members on a map (approximate ~1 km pins for privacy)	✅
G2	Filter listings by city and by distance radius; sort by nearest	✅
G3	Suggested nearby swaps ranked by value, distance and fit	✅

ID	Requirement	Status
G4	Distance on every card for logged-in members	✅

6.7 Admin module

ID	Requirement	Status
A1	Manage users: search, filter, suspend (with reason, closes open requests), reactivate	✅
A2	Manage listings: search, filter (reported/status), remove with reason, restore	✅
A3	Monitor swap activity with filters	✅
A4	Review reports: remove listing, suspend member, warn, dismiss	✅
A5	Resolve disputes: reverse (release items) or mark completed, with a note to both members	✅
A6	Analytics: KPIs, 12-week activity, funnel, categories, cities, top swappers, impact	✅
A7	Reports: CSV export of users, items, swaps	✅

7. Non-functional requirements

Area	Implementation
Secure authentication	bcrypt (cost 10) · JWT in httpOnly, SameSite=Lax, Secure (prod) cookie · suspended users rejected on every request · auth rate limit 30 / 15 min
Input validation	Zod schemas on every write endpoint; mirrored client-side validation with inline errors
Injection & XSS	`$`/dotted keys stripped from input; regex-escaped search; React escaping; Helmet CSP; CSV formula-injection guard
Privacy	Exact coordinates never returned — only area name, distance and ~1 km rounded map pins; phone numbers private; chats only visible to participants
Performance	2dsphere + compound indexes; `$geoNear` aggregation with facet pagination; code-split routes; immutable image caching; gzip
Responsive	Mobile-first Tailwind layout, tested at 390 px and 1440 px
Scalability	Stateless API (JWT), works as serverless functions or a long-running server; connection pooling; images stored in MongoDB for zero extra services
Accessibility	Semantic landmarks, labelled controls, focus rings, `aria-pressed` on toggles, reduced-motion support, colour + text for every status
Reliability	Atomic reservation on accept; idempotent auto-seed with a lock for concurrent cold starts

8. Swap value algorithm

```
reference retail = MRP (if given)  or  category reference price × brand price index
swap value       = reference retail × brand resale rate × condition multiplier × age
```

```
multiplier
                    (rounded to ₹50; owner may adjust ±25%)
```

Brand tier	Example brands	Price index	Resale rate
Luxury / Designer	Gucci, Sabyasachi, Anita Dongre	×12	45%
Premium	Levi's, Nike, Fabindia, Raymond	×2	32%
High-street	Zara, H&M, Uniqlo, Biba, W	×1	22%
Value	Roadster, Max, Decathlon	×0.7	20%
Local / Unbranded	Handmade, market finds	×0.6	20%

Condition: New with tags ×1.45 · Like new ×1.2 · Good ×1.0 · Fair ×0.7. Age: < 1 yr ×1 · 1–2 yrs ×0.92 · 2–4 yrs ×0.82 · 4+ yrs ×0.72. Resale rates reflect published resale-to-retail ratios (fast fashion ≈ 17–23%, premium ≈ 27–45%, luxury ≈ 40–60%).

Fairness = min(A, B) ÷ max(A, B) → ≥ 90% *Excellent* · ≥ 75% *Fair* · ≥ 60% *Slightly uneven* · else *Uneven*.

Fair matches (item page): all single items and pairs from the viewer's closet, ranked by fairness (≥ 60%).

Nearby suggestions: candidates within the radius (via `$geoNear`), filtered to the genders the member lists, scored `0.5·fairness + 0.3·proximity + 0.2·fit`, where fit = wishlist category (0.4) + owner wants your category (0.3) + in your size (0.3).

Courier estimate: Local (≤ 50 km) ₹70, Regional (≤ 500 km) ₹95, National ₹125 for the first 0.5 kg, plus ₹35–55 per extra 0.5 kg (based on category weights).

9. Impact model

Each completed swap adds the received garments' typical weight (e.g. T-shirt 0.2 kg, jeans 0.6 kg, lehenga 1.5 kg) to a member's "kg kept in use". CO_2e avoided = 25 kg per kg reused (INTEXTER–UPC, 2022); water saved ≈ 13,500 L per kg (WWF: ~2,700 L per 0.2 kg cotton T-shirt). Figures are labelled as estimates.

10. User flows

Member: Register → set neighbourhood → list items (calculator suggests value) → browse / nearby / suggestions → open item → see fair matches → request swap → negotiate (counter-offers + chat) → agreement (items reserved) → plan meetup or add courier tracking → both confirm receipt → rate → impact updates.

Admin: Log in → analytics overview → review reports (remove / suspend / warn / dismiss) → resolve disputes → moderate listings and users → export CSV reports.

Swap state machine

```
pending ──accept──▶ accepted ──plan/ship──▶ in_progress ──both received──▶ completed
   │  ▲                │                        │
counter│               cancel (nothing shipped)  dispute ──admin──▶ completed |
cancelled
   ▼  │                ▼                        ▼
countered            cancelled                disputed
   │
reject/withdraw ──▶ rejected / cancelled
```

11. Data model

Collection	Key fields
users	name, email (unique), password (hash), role, status, phone (private), bio, location {city, area, point (2dsphere)}, sizes, preferences, favorites, rating {avg, count}, stats {swapsCompleted, kgDiverted}, lastActiveAt
items	owner, title, description, category, gender, size, brand, brandTier, condition, colour, material, age, originalPrice, estimatedValue, swapValue, lookingFor, images, location (2dsphere), status, moderation, views, favoritesCount
swaps	requester, recipient, offeredItems, requestedItems, offers[] (each round with values & fairness), awaiting, status, distanceKm, fulfillment {method, meetup, shipments, estimate}, received, ratings, dispute, timeline, conversation
conversations	participants, key (pair), item, swap, lastMessage, unread (per user)
messages	conversation, sender, type (text / offer / system), text, swap
images	owner, contentType, size, data (binary)
reports	reporter, targetType, item, user, reason, details, status, handledBy, note
notifications	user, type, title, body, link, read

Sample clothing data fields required by the brief (item ID, type, brand, size, condition, estimated value, location) and sample user data (name, contact, location, swap history) are all present. The seed contains **24 members in 10 Indian cities, 65 live listings + 24 historical items, 33 swaps across every status, 92 messages, reports and a dispute**.

12. API overview

Area	Endpoints
Auth	`POST /api/auth/register` · `POST /api/auth/login` · `POST /api/auth/logout` · `GET /api/auth/me`
Items	`GET /api/items` (q, category, gender, size, condition, brand, city, min/maxValue, lat/lng, radiusKm, sort, page) · `GET/PATCH/DELETE /api/items/:id` · `POST /api/items` · `POST /api/items/:id/favorite`

Area	Endpoints
Swaps	`POST /api/swaps` · `GET /api/swaps` · `GET /api/swaps/:id` · `POST /api/swaps/:id/{respond,cancel,fulfillment,ship,received,dispute,rate}`
Chat	`GET/POST /api/chat/conversations` · `GET /api/chat/conversations/:id` · `POST …/messages` · `POST …/read` · `GET /api/chat/unread`
Users	`PATCH /api/users/me` · `GET /api/users/me/{items,favorites,suggestions,dashboard}` · `GET /api/users/nearby` · `GET /api/users/:id`
Other	`POST /api/images` · `GET /api/images/:id` · `GET /api/meta` · `GET /api/stats` · `POST /api/valuation/estimate` · `GET /api/notifications` · `POST /api/reports`
Admin	`GET /api/admin/overview` · `GET/PATCH /api/admin/users` · `GET/PATCH /api/admin/items` · `GET /api/admin/swaps` · `POST /api/admin/swaps/:id/resolve` · `GET/PATCH /api/admin/reports` · `GET /api/admin/export/:type`

13. KPIs (tracked live in the admin panel)

KPI (brief)	Definition in ReThread
Number of clothing listings	Listings ever published (available + reserved + swapped); available shown separately
Number of successful swaps	Swaps with status *completed*
User engagement rate	Members active in the last 30 days who listed, swapped or chatted ÷ all members
Swap request conversion rate	Requests that reached agreement (accepted, in progress, completed, disputed) ÷ all requests; completion rate shown alongside
Active users	Members with activity in the last 30 days
Impact (extra)	kg of textiles kept in use, kg CO_2e avoided

14. Pages (brief requires 6–8; ReThread ships 13)

Landing · Login/Register · Clothing listings (Browse) · Item detail · List/Edit item · Swap request · Swaps & swap detail · Chat · User dashboard · Nearby map · Value calculator · Profile & Settings · Admin panel.

15. Testing

- **Automated:** 11 end-to-end API tests (`npm test`) on an in-memory MongoDB with the full seed — auth & validation, injection attempt, role guards, listing bounds, geo search ordering, complete swap lifecycle with counter-offer and auto-cancellation of competing requests, chat privacy, suggestions, moderation, dispute resolution, CSV export, suspension.
- **UI:** every page rendered in Chromium at 1440 px and 390 px with zero console errors; screenshots in the submission.

- **Manual checklist:** register → list → request → counter → accept → meetup → receive → rate; admin removes a reported listing and resolves a dispute.

16. Assumptions & constraints

- Members upload accurate information; quality depends on members → reports, ratings and disputes mitigate.
- Swap fairness is ultimately agreed by members → the calculator is guidance, not enforcement.
- Courier costs are borne by members → estimates are shown before choosing courier; meetups are encouraged.
- Location uses curated neighbourhoods + browser GPS (no paid geocoding).

17. Future enhancements

AI-based swap recommendations and photo-based condition checks · native mobile app with push notifications · in-app courier booking via Shiprocket/Delhivery APIs · sustainability leaderboard and badges · community fashion groups and swap events · multi-party swap chains · Hindi and regional-language UI.

18. Deliverables

Deliverable	Where
Fully functional Clothing Exchange Marketplace	Source code (this repository)
Listing & browsing, swap request & negotiation, location matching, admin panel	Implemented — see §6
Live deployed application link	**https://rethread-ten.vercel.app** (Vercel, Mumbai + MongoDB Atlas)
Complete PRD document	This document

Sources

Fashion for Good, *Wealth in Waste: India's Potential to Lead Circular Textile Sourcing* (2022) · INTEXTER–UPC, *Reusing 1 kg of clothing saves 25 kg of CO_2* (2022) · Closo, *Resale Price vs Original: Ratios by Category* (2026) · YourStory, *This For That* (2019) · Unified Mentor project brief.